

# Excel Lookup Functions Documentation

## Module: VLOOKUP & HLOOKUP & XLOOKUP in Excel

Using Microsoft Excel

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### Learning Objectives

By the end of this session, learners will be able to:

- Understand lookup functions
  - Use VLOOKUP for vertical data retrieval
  - Use HLOOKUP for horizontal data retrieval
  - Apply lookup functions in real-time business scenarios
  - Handle exact and approximate matches
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### ◆ Introduction to Lookup Functions

Lookup functions are used to:

- ✓ Search data
  - ✓ Retrieve matching values
  - ✓ Automate reporting
  - ✓ Reduce manual work
- 

### Real-Time Example

Suppose you have:

- Employee ID
- Employee Name
- Salary

Instead of searching manually,  
Excel can automatically fetch details using lookup functions.

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## A. VLOOKUP for Vertical Data Retrieval

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### 1 What is VLOOKUP?

#### ◆ Meaning

VLOOKUP = Vertical Lookup

Used when:

👉 Data is arranged vertically (top to bottom)

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#### ◆ Syntax

None

```
=VLOOKUP(lookup_value, table_array, col_index_num, [range_lookup])
```

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### Understanding Parameters

| Parameter    | Meaning         |
|--------------|-----------------|
| lookup_value | Value to search |

|               |                                 |
|---------------|---------------------------------|
| table_array   | Entire table                    |
| col_index_num | Column number to return         |
| range_lookup  | Exact/FALSE or Approximate/TRUE |

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## 2 Basic Example

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### Employee Dataset

| Emp_ID | Name  | Department | Salary |
|--------|-------|------------|--------|
| 101    | Ravi  | HR         | 25000  |
| 102    | Anita | IT         | 30000  |
| 103    | John  | Finance    | 28000  |

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### ◆ Goal

Find employee name using Emp\_ID.

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### ◆ Formula

None

`=VLOOKUP(102, A2:D4, 2, FALSE)`

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## Explanation

Part

Meaning

102      Search value  
A2:D4    Table  
2        Return 2nd column  
FALSE    Exact match

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## ◆ Result

None  
Anita

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## Important Rule

VLOOKUP searches only in:

☒ First column of table

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## Wrong Example

If Emp\_ID is not first column:

None  
VLOOKUP will not work properly

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## Exact Match vs Approximate Match

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### ◆ Exact Match (FALSE)

Used for:

- Employee IDs
- Product IDs
- Phone numbers

Formula:

None

```
=VLOOKUP(101, A2:D10, 2, FALSE)
```

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## ◆ Approximate Match (TRUE)

Used for:

- Grades
- Commission slabs
- Tax calculations

Formula:

None

```
=VLOOKUP(75, A2:B5, 2, TRUE)
```

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## Example: Student Grades

| Marks | Grade |
|-------|-------|
|-------|-------|

|   |   |
|---|---|
| 0 | C |
|---|---|

|    |   |
|----|---|
| 40 | B |
|----|---|

|    |   |
|----|---|
| 75 | A |
|----|---|

Formula:

None

=VLOOKUP(80, A2:B4, 2, TRUE)

Result:

None

A

## 4 Real-Time Business Scenario

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### ◆ Hospital Example

| Patient_ID | Name | Bill |
|------------|------|------|
| 101        | Ravi | 5000 |

Goal:

Fetch bill amount using Patient\_ID.

Formula:

None

=VLOOKUP(101, A2:C10, 3, FALSE)

## 5 Common Errors in VLOOKUP

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### ◆ #N/A Error

Reason:  
Value not found.

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## ◆ Wrong Column Index

Example:

None  
`=VLOOKUP(101, A2:D5, 10, FALSE)`

✗ Error because column 10 doesn't exist.

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## ◆ Missing FALSE

If FALSE not used:  
Excel may return incorrect results.

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## 🔥 Best Practice

Always use:

None  
`FALSE`

for IDs and names.

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## 🧪 Practice Activity

Create:

| Product_ID | Product | Price |
|------------|---------|-------|
|------------|---------|-------|

Use VLOOKUP to:

- Find Product Name
  - Find Product Price
- 

## Advanced Example

### Using Cell Reference

Instead of hardcoding:

None  
`=VLOOKUP(102, A2:D5, 2, FALSE)`

Use:

None  
`=VLOOKUP(F2, A2:D5, 2, FALSE)`

Where:

None  
`F2 contains Employee ID`

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## Dynamic Lookup System

User enters:



None

Employee ID

Excel automatically returns:

- Name
- Salary
- Department

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## B. HLOOKUP for Horizontal Data Retrieval

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### 1 What is HLOOKUP?

#### ◆ Meaning

HLOOKUP = Horizontal Lookup

Used when:

👉 Data is arranged horizontally (left to right)

#### ◆ Syntax

None

```
=HLOOKUP(lookup_value, table_array, row_index_num, [range_lookup])
```

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### Difference Between VLOOKUP & HLOOKUP

| Function | Searches     |
|----------|--------------|
| VLOOKUP  | Vertically   |
| HLOOKUP  | Horizontally |

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## 2 Example

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|         | Jan  | Feb  | Mar  |
|---------|------|------|------|
| Revenue | 5000 | 7000 | 8000 |

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### ◆ Goal

Find February revenue.

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### ◆ Formula

None

`=HLOOKUP("Feb", A1:D2, 2, FALSE)`

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### ◆ Result

None

`7000`

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## Explanation

| Part  | Meaning        |
|-------|----------------|
| "Feb" | Search value   |
| A1:D2 | Table          |
| 2     | Return 2nd row |
| FALSE | Exact match    |

---



## Real-Time Example

### Monthly Sales Dashboard

| Month | Jan   | Feb   | Mar   |
|-------|-------|-------|-------|
| Sales | 50000 | 60000 | 55000 |

Use HLOOKUP to retrieve:

- Monthly sales
  - Monthly targets
- 

## 3 VLOOKUP vs HLOOKUP

| Feature      | VLOOKUP      | HLOOKUP     |
|--------------|--------------|-------------|
| Direction    | Vertical     | Horizontal  |
| Searches In  | First column | First row   |
| Common Usage | Most used    | Rarely used |

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## Industry Usage

**VLOOKUP** is heavily used in:

- HR
  - Finance
  - Healthcare
  - Sales
  - Data Analysis
- 

**HLOOKUP** is used in:

- Monthly reports
  - Horizontal dashboards
  - Summary sheets
- 

## Final Practice Exercise

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### Dataset 1 (VLOOKUP)

| Emp_ID | Name  | Salary |
|--------|-------|--------|
| 101    | Ravi  | 25000  |
| 102    | Anita | 30000  |

Tasks:

- ☒ Fetch employee name
  - ☒ Fetch salary
- 

### Dataset 2 (HLOOKUP)

|         | Q1    | Q2    | Q3    |
|---------|-------|-------|-------|
| Revenue | 50000 | 60000 | 70000 |

Tasks:

- ☒ Retrieve Q2 revenue
- ☒ Retrieve Q3 revenue

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## End of Session Assignment

Build:

### Employee Information System

User enters:

None

Employee ID

Excel should display:

- Name
- Department
- Salary

Using:

- ☒ VLOOKUP
  - ☒ IFERROR (optional advanced)
-

# C. XLOOKUP – Modern Lookup Function (Recommended)

## 1 What is XLOOKUP?

### ♦ Meaning

**XLOOKUP = Extended Lookup**

It is the latest lookup function introduced by Microsoft Excel that overcomes the limitations of **VLOOKUP** and **HLOOKUP**.

Unlike VLOOKUP, XLOOKUP can:

-  Search left to right
-  Search right to left
-  Search vertically and horizontally
-  Return custom messages if data is not found
-  Eliminate the need to count column numbers

## 2 Syntax

None

```
=XLOOKUP(lookup_value, lookup_array, return_array, [if_not_found],  
[match_mode], [search_mode])
```

## Understanding Parameters

| Parameter    | Meaning                                    |
|--------------|--------------------------------------------|
| lookup_value | Value to search                            |
| lookup_array | Column or row where Excel searches         |
| return_array | Column or row from which value is returned |
| if_not_found | Message if value is not found (Optional)   |

match\_mode    Exact or approximate match (Optional)

search\_mod    Search direction (Optional)  
e

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## 3 Basic Example

### Employee Dataset

| Emp_ID | Name  | Department | Salary |
|--------|-------|------------|--------|
| 101    | Ravi  | HR         | 25000  |
| 102    | Anita | IT         | 30000  |
| 103    | John  | Finance    | 28000  |

---

### Goal

Find Employee Name using Employee ID.

### Formula

None

`=XLOOKUP(102, A2:A4, B2:B4)`

### Result

None

`Anita`

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### Explanation

| Part | Meaning         |
|------|-----------------|
| 102  | Value to search |

A2:A4    Search column

B2:B4    Return column

---

## 4 Fetch Salary

### Formula

None

`=XLOOKUP(102, A2:A4, D2:D4)`

### Result

None

`30000`

---

## 5 Using Cell Reference

Instead of typing the Employee ID directly:

None

`=XLOOKUP(F2, A2:A10, B2:B10)`

Where:

- **F2** contains Employee ID entered by the user.

This creates a **dynamic lookup system**.

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## 6 Returning Multiple Values

Suppose the user enters Employee ID in **F2**.



## Employee Name

None

```
=XLOOKUP(F2, A2:A10, B2:B10)
```

## Department

None

```
=XLOOKUP(F2, A2:A10, C2:C10)
```

## Salary

None

```
=XLOOKUP(F2, A2:A10, D2:D10)
```

Excel automatically displays all employee details.

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# 7 Search Left (Major Advantage)

## Dataset

| Name  | Emp_ID |
|-------|--------|
| Ravi  | 101    |
| Anita | 102    |
| John  | 103    |

Suppose you want to find the **Employee ID** using the employee name.

## Formula

None

```
=XLOOKUP("Anita", A2:A4, B2:B4)
```

## Result

None

102

**VLOOKUP** cannot perform this because it only searches from left to right.

---

## 8 If Value Not Found

Instead of showing **#N/A**, XLOOKUP can display a custom message.

### Formula

None

```
=XLOOKUP(105,A2:A4,B2:B4,"Employee Not Found")
```

### Result

None

Employee Not Found

This improves user experience and makes reports cleaner.

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## 9 Approximate Match

### Grade Dataset

| Marks | Grade |
|-------|-------|
|-------|-------|

|   |   |
|---|---|
| 0 | C |
|---|---|

|    |   |
|----|---|
| 40 | B |
|----|---|

|    |   |
|----|---|
| 75 | A |
|----|---|

### Formula

None

```
=XLOOKUP(80,A2:A4,B2:B4,"Not Found",-1)
```

## Result

None

A

## Match Mode Values

| Match Mode | Meaning                           |
|------------|-----------------------------------|
| 0          | Exact Match (Default)             |
| -1         | Exact match or next smaller value |
| 1          | Exact match or next larger value  |
| 2          | Wildcard Match                    |

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## 10 Search from Bottom to Top

Suppose duplicate Employee IDs exist and you want the latest entry.

## Formula

None

```
=XLOOKUP(101,A2:A20,D2:D20,"Not Found",0,-1)
```

## Result

Returns the **last occurrence** of Employee ID 101.

---

## 11 Real-Time Business Scenario

## HR Employee Management

Employee enters:

- Employee ID

Excel automatically displays:

- Employee Name
- Department
- Salary
- Designation

Using:

None

```
=XLOOKUP(F2, A2:A100, B2:B100)
```

This is widely used in:

- HR
- Banking
- Finance
- Healthcare
- Sales Reporting
- Inventory Management

---

## 12 Common Errors

### Incorrect Lookup Range

 Wrong

None

```
=XLOOKUP(102, A2:A4, D2:D20)
```

The lookup and return arrays must have the **same number of rows or columns**.

---

## Value Not Found

Without the optional argument:

None

```
=XLOOKUP(110, A2:A4, B2:B4)
```

Result:

None

```
#N/A
```

Better approach:

None

```
=XLOOKUP(110, A2:A4, B2:B4, "Employee Not Found")
```

## Advantages of XLOOKUP over VLOOKUP & HLOOKUP

| Feature                               | VLOOKUP              | HLOOKUP              | XLOOKUP |
|---------------------------------------|----------------------|----------------------|---------|
| Vertical Lookup                       | ✓                    | ✗                    | ✓       |
| Horizontal Lookup                     | ✗                    | ✓                    | ✓       |
| Search Left                           | ✗                    | ✗                    | ✓       |
| Search Right                          | ✓                    | ✓                    | ✓       |
| No Column Number Required             | ✗                    | ✗                    | ✓       |
| Custom Not Found Message              | ✗ (requires IFERROR) | ✗ (requires IFERROR) | ✓       |
| Returns Multiple Types of Data Easily | ✗                    | ✗                    | ✓       |

Recommended by Microsoft

No

No



## Practice Activity

### Employee Dataset

| Emp_ID | Name  | Department | Salary |
|--------|-------|------------|--------|
| 101    | Ravi  | HR         | 25000  |
| 102    | Anita | IT         | 30000  |
| 103    | John  | Finance    | 28000  |

### Tasks

1. Retrieve Employee Name.
2. Retrieve Department.
3. Retrieve Salary.
4. Display "**Employee Not Found**" if the ID does not exist.
5. Find Employee ID using Employee Name (left lookup).